

Multiple Choice (10 marks)

1. Which among the following is the least strong security encryption standard?
 - (a) WEP
 - (b) WPA
 - (c) WPA 2
 - (d) WPA 3
2. _____ is also a part of darknet that is employed for transferring files anonymously.
 - (a) Freenet
 - (b) ARPANET
 - (c) Stuxnet
 - (d) Internet
3. When does a buffer overflow occur, generally speaking?
 - (a) when writing to a pointer that has been freed
 - (b) when copying a buffer from the stack to the heap
 - (c) when a pointer is used to access memory not allocated to it
 - (d) when the program notices a buffer has filled up, and so starts to reject requests
4. Public key encryption is advantageous over Symmetric key Cryptography because of
 - (a) Speed
 - (b) Space
 - (c) Key exchange
 - (d) Key length
5. The _____ can cost you money, by sending text messages from your mobile phone numbers.
 - (a) IM - Trojans
 - (b) Backdoor Trojans
 - (c) SMS Trojan
 - (d) Ransom Trojan

True / False (10 marks)

Answer True or False

6. True or False: A MAC with a fixed tag length can still be secure if the underlying algorithm is robust and employs a strong key management strategy.
7. True or False: WPA relies on AES encryption exclusively for securing wireless communications.
8. True or False: EXE restarts the STP solver and retries the query indefinitely upon a timeout on a constraint query.
9. True or False: Mutation-based fuzzing only applies to file-based inputs and is not effective for network protocol testing.
10. True or False: Whitebox fuzzers are designed to explore all paths in a program, ensuring coverage of every line of code.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to count repeated items of a tuple.
12. Write a python function to remove spaces from a given string.

End of Paper